

ORIGINAL ARTICLE

Far-Field Magnetic Sensing on Soft Origami Actuator for Spatial Multidimensional Movement and Force Perception

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Abstract

Soft robots exhibit exceptional flexibility and adaptability, enabling them to dynamically adjust their body shapes in unstructured environments for a wide range of applications. This motivates an extensive investigation of soft sensing techniques to accommodate such versatility. Magnetic sensing mechanisms present promising adaptability for soft robots due to their ease of integration and extensive detection range features. In this study, we explore the potential of utilizing far-field magnetic sensing in combination with a soft actuator model-based approach for multimodal perception. We introduce a Soft Farfield Magnetic Origami design, which incorporates a concise Hall sensory array into soft pneumatic origami actuators. The Hall sensory array is utilized to track the unique distal position of the soft actuator. This facilitates the further retrieval of spatial multidimensional movements, including linear and omnidirectional bending motions, as well as interactive forces. This multimodal sensing capability is supported by the modeled relationships between the desired sensing modalities and the measurable set of soft origami actuators, in terms of distal position and pressure. Our proposed approach showcases accurate spatial kinematic perception with a root-mean-square deviation of 0.36 mm in length, 0.02 rad in angle, and an interactive force variation detection with a root-mean-square deviation of 0.89 N. This comprehensive methodology from concept, modeling, design, and fabrication, to validation, facilitates position feedback control and interactive force tuning in soft robotic systems.

Keywords: soft robotics, soft actuator, soft sensor, far-field magnetic sensing, origami

Introduction

Spatial movement and force perception, which represent both the state of the robot and the environment in interactions, are essential for feedback control and decision-making in many manipulation and exploration tasks.¹ In recent years, soft robotics has rapidly garnered interest in the field of robotics. Benefiting from unique soft materials and structures, soft robots are endowed with high performance to fulfill numerous tasks, making them ideal for a wide range of applications, such as manipulation, exploration, and navigation.^{2–9} This motivates extensive research into soft sensing techniques to accommodate such versatility. However,

owing to the inherent compliance, soft robots are presented with challenges in measuring the complex body deformations for multimodal perception.^{10,11}

Over the past decades, the deformation characteristics of soft robots have been extensively explored using mathematical analysis, simulations, and experimental observations.^{11–13} Both static modeling and dynamic modeling of soft actuators on spatial kinematics and kinetics have been studied, characterizing the continuous and large strain from the soft materials.^{14–19} Based on these characteristics, many efforts have been made on strain-based sensing by locally embedding soft strain sensors into soft bodies, with implementation of various sensing mechanisms.^{20–24} The utilization of

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soft materials as substrates enables soft strain sensors to seamlessly conform to soft body shape changes, maintaining compliance and adaptability in soft robots. Leveraging these characteristics, significant progress has been made in the optimization of soft strain sensor design and integration.^{25–32} Aided by model-based or learning-based techniques, these studies have achieved accurate measurement of soft robot movements and forces. Alternatively, more recently, no-strain-based sensing methods^{33–35} have been developed, which leverage optical and pressure information instead of conventional strain to retrieve the multimodal sensing information. These innovative approaches channel optical lights within the cavity chamber to detect the distal position, effectively eliminating the need for soft material substrates on soft strain sensors, thereby offering simplification for soft robotic design and fabrication. However, this line-of-sight access principle also introduces limitations on the range of motion and flexibility of soft robots in complex environments. In contrast, the far-field magnetic sensing mechanism offers a better alternative with its ability to penetrate obstacles. Pioneering efforts in utilizing this mechanism have been dedicated,³⁶ showcasing high sensing performance in spatial movement measurement and high adaptability for some extreme environments. This holds promising prospects for enhanced sensing capabilities, ultimately advancing the overall control performance of soft robotics for various applications.

In this article, we explore the potential of utilizing far-field magnetic sensing in combination with a soft actuator model-based approach for retrieving multidimensional movement and force information. This approach is based on the measurement of no-strain measurable set, which includes the distal position obtained using a triangularly arranged Hall sensor array, as well as the actuator pressure. By integrating this array into a soft pneumatic origami actuator, we introduce a Soft Far-Field Magnetic Origami (SoMO) design with a concise design configuration for constructing sensor-integrated soft robots. Detailed models are established to elucidate the relationships between the no-strain measurable set of soft origami actuators (distal position and pressure) and desired sensing modalities. This facilitates the retrieval of

spatial multidimensional movements (extension, contraction, and omnidirectional bending) and interactive forces (axial force in linear and bending movements). A position feedback controller and interactive force tuning strategy are implemented to the SoMO actuator and joint system, showcasing the preliminary control performance of the proposed approach.

The main contributions of this work include the following: (1) Utilization of far-field magnetic sensing mechanism with a soft actuator model-based sensing approach for retrieving multidimensional movement and force information, (2) SoMO design with concise sensor configuration and being supported by models that elucidate the relationships between desired sensing modalities and measurable variables, and (3) spatial position feedback control and interactive force tuning strategy in the soft robotic system, validating the proposed approach with a complete methodology from concept, modeling, design, and fabrication to system development.

This article is structured as follows: The section design principle and modeling introduces the modeling of the proposed approach and the design principles of the Hall array. The section workspace verification section provides a preliminary verification of the conceptual design. In verification on spatial movement perception section, we present the verification of spatial multidimensional movement estimation, including extension, contraction, and omnidirectional bending. The section verification on interactive force perception discusses the perception of interactive forces. Finally, demonstrations of the SoMO actuator and joint robotic system are presented in the section demonstration on soft robotic system.

Design Principle and Modeling

Concept

The proposed approach is validated using the SoMO actuator, which comprises an origami chamber sandwiched between a triangularly arranged Hall sensor array and a permanent magnet, as illustrated in Figure 1a. A far-field magnetic Hall sensing mechanism is used to capture the distal position and soft body shapes, as shown in Figure 1b. When

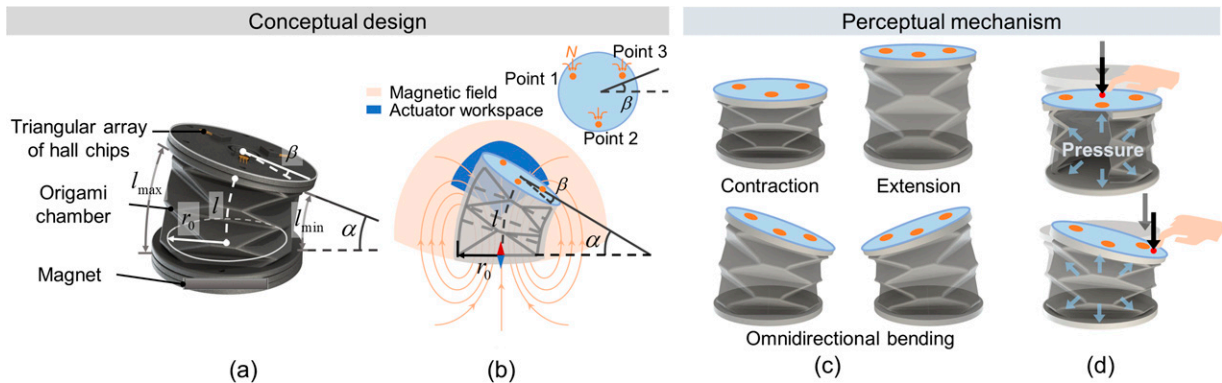


FIG. 1. Concept of the proposed sensing approach. (a) Origami chamber sandwiched between a triangularly arranged Hall sensor array and a permanent magnet based on the far-field magnetic sensing mechanism. (b) The conceptual Soft Far-Field Magnetic Origami (SoMO) design features an actuator workspace (represented by the blue area, which indicates the trajectory of the distal plate center) encompassed by the magnetic field (shown in orange to signify the magnetic field's coverage). (c) Mechanisms of sensing the spatial multidimensional movements, and (d) forces during interactions.

subjected to external forces or internal actuation, the Hall sensory array (labeled at Points 1, 2, and 3 in Fig. 1b) detects variations in the distal position. This array requires a minimum of three triangularly arranged Hall sensors to determine the unique distal position, which further enables the retrieval of spatial multidimensional movements based on the commonly known piecewise constant curvature theory^{11,12} for soft actuators (Fig. 1c). This information is subsequently utilized to estimate interactive forces based on the intrinsic relationships between spatial kinematics and forces in SoMO actuators (Fig. 1d).

Spatial kinematic model and sensory array design

The distal plate can be geometrically defined by three points located at the vertices of the triangle. Upon acquiring the distal position, spatial multidimensional movements can be measured by characterizing the soft body shape based on piecewise constant curvature theory. The actuator body shape could be fully defined by (α, β, l) in the actuator coordinate frame, or by coordinates of points (H_1, H_2, H_3) in the Cartesian coordinate frame. Here, α is the bending angle of actuator. β is the orientation of distal plate from the initial x_0 -axis in the distal plate to the bending position. l is the central length of actuator. Points $H_{i,i=1,2,3}$ are the positions inserting Hall sensors, as illustrated in Figure 2a. In the actuator coordinate frame, α , β , and l have the following relations^{11,12}:

$$\begin{cases} l_{i,i=1,2,3,\dots,n} = l - \alpha r \cos(\theta_i - \beta) \\ l = (l_1 + l_2 + \dots + l_n)/n \end{cases} \quad (1)$$

where θ_i is the mounting angle from the axis x_0 to the Hall sensor, l_i the length of sensor lines in parallel with the central line, and r is the distance of Hall sensor to the center of distal plate as illustrated in Figure 2b. In Equation (1), to determine the variables of α , β , and l , a regular polygon, with n vertices centered around the distal plate can be used to define the shape of the soft body. The simplest configuration is an equilateral triangle, where $n = 3$ according to this equation. The variable r , which represents the dimensions of the sensory array, can be customized to accommodate the design of various soft actuators. For instance, when two soft actuators have distal plates with small and large radii, the value of r can be customized for each distal plate or maintained the

same for both. Therefore, the transformation from l_1, l_2, l_3 to the actuator coordinate frame satisfies the following:

$$\begin{cases} l = (l_1 + l_2 + l_3)/3 \\ \beta = \text{atan} 2(\cos \theta_1 - n_0 \cos \theta_2, n_0 \sin \theta_2 - \sin \theta_1) \\ \alpha = (l - l_1)/(r \cos(\theta_1 - \beta)) \end{cases} \quad (2)$$

where $n_0 = (l - l_1)/(l - l_2)$. Equation (2) establishes the relationship between the body shape in the actuator coordinate frame (α, β, l) and variables of (l_1, l_2, l_3) . It plays a crucial role in the transformation between the Cartesian coordinate frame and the actuator coordinate frame. Particularly, parameters $\alpha = 0, \beta = 0$ in the linear actuation state. As the central line is approximated to be with constant curvature, coordinates of center of distal plate O_0 geometrically are as follows:

$$O_0 \left(\frac{l(1 - \cos \alpha) \cos \beta}{\alpha}, \frac{l(1 - \cos \alpha) \sin \beta}{\alpha}, \frac{l \sin \alpha}{\alpha} \right). \quad (3)$$

The coordinates of Hall sensors $H_{i,i=1,2,3}$ (x_i, y_i, z_i) are

$$H_i \left(r \cos \theta_i + \frac{l_i(1 - \cos \alpha) \cos \beta}{\alpha}, r \sin \theta_i + \frac{l_i(1 - \cos \alpha) \sin \beta}{\alpha}, \frac{l_i \sin \alpha}{\alpha} \right). \quad (4)$$

Therefore, by substituting Equations (1) into (4), we have completed the transformation from the actuator coordinate frame to the Cartesian coordinate frame as follows:

$$\begin{cases} x_i = r \cos \theta_i + [l - \alpha r \cos(\theta_i - \beta)](1 - \cos \alpha) \cos \beta / \alpha \\ y_i = r \sin \theta_i + [l - \alpha r \cos(\theta_i - \beta)](1 - \cos \alpha) \sin \beta / \alpha \\ z_i = [l - \alpha r \cos(\theta_i - \beta)] \sin \alpha / \alpha \end{cases} \quad (5)$$

According to Equations (2) and (5), the key for defining the spatial kinematics in the actuator coordinate frame and the Cartesian coordinate frame is to obtain the lengths of l_1, l_2, l_3 , which could be decoded by the signals of three evenly distributed Hall sensors (h_1, h_2, h_3). The output voltages of Hall sensors follow a linear relation to the magnetic flux ϕ , satisfying

$$h_i = h_0 + c\phi. \quad (6)$$

where $i = 1, 2, 3$, c is a constant related to the specification of Hall sensor and can be obtained experimentally, and h_0 is the initial voltage specified for the Hall sensors:

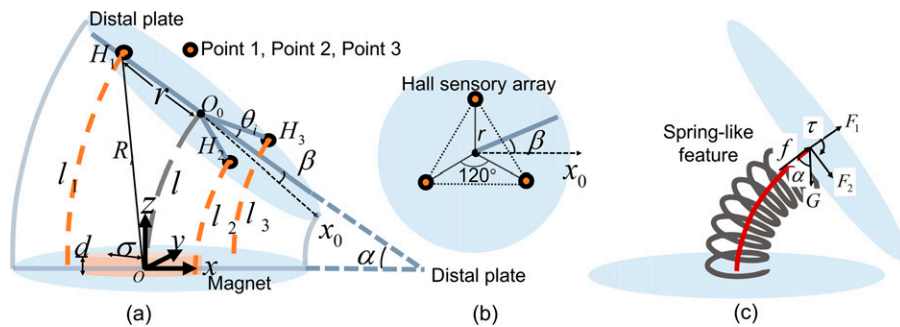


FIG. 2. (a) Diagram of spatial kinematic illustration. The yellow dotted lines represent the curve lengths decoded by Hall sensors. The gray dotted line indicates the central line of the soft actuator. The blue plates correspond to the interactive plane and mounting plane, respectively. (b) Illustration of the Hall sensor array. (c) Modeling representation of interactive forces, with the spring symbolizing the spring-like feature.

$$\phi = \mathbf{B} \cdot \mathbf{n}, \quad (7)$$

where $\mathbf{B}$ is the magnetic flux density, and $\mathbf{n}$ is the unit normal vector that connects to the orientation of distal plate. In this design,

$$\mathbf{n} = (\sin \alpha \cos \beta, \sin \alpha \sin \beta, \cos \alpha)^T. \quad (8)$$

Based on the well-documented studies,^{37,38} the magnetic density for a spatial point $\mathbf{H}_i (x_i, y_i, z_i)$ relative to magnet position $O (0, 0, 0)$ satisfies

$$\mathbf{B} = \frac{\mu_T \mu_0 m_T}{4\pi} \left(\frac{3(\mathbf{H}_0 \cdot \mathbf{H}_i) \mathbf{H}_i}{R_i^5} - \frac{\mathbf{H}_0}{R_i^3} \right), \quad (9)$$

where $\mathbf{H}_0$ is the normalized vector of magnetic intensity of magnet, and $\mathbf{H}_0 = (0, 0, 1)^T$ for this conceptual design. $\mu_T = 1$ is the relative permeability of the air. $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ is the air magnetic permeability. $R_i = (x_i^2 + y_i^2 + z_i^2)^{1/2}$ is the distance to the magnet's center. m_T defines the magnetic intensity of magnet, which is constant for a permanent magnet,

$$m_T = \pi \sigma^2 d m_0, \quad (10)$$

where σ is the radius of a cylindrical magnet, d is the thickness of cylindrical magnet, m_0 is the magnetization strength $m_0 = 1.032 \times 10^6 \text{ A/m}$ for an Nd-Fe-B magnet. Therefore, combining Equations (7)–(10) and substituting them into Equation (6),

$$h_{i=1,2,3\dots} = h_0 + \frac{c\mu_T\mu_0m_T}{4\pi} \left(\frac{3(\mathbf{H}_0 \cdot \mathbf{H}_i) \mathbf{H}_i}{R_i^5} - \frac{\mathbf{H}_0}{R_i^3} \right) \cdot \mathbf{n}. \quad (11)$$

Substituting Equations (2) and (5) into (11), we have

$$h_i = h_0 + \frac{c\mu_T\mu_0m_T \cos \alpha}{4\pi} \left[\frac{3z_i^2 - (x_i^2 + y_i^2 + z_i^2)^2}{(x_i^2 + y_i^2 + z_i^2)^{5/2}} \right], \quad (12)$$

where x_i, y_i, z_i refer to Equation (6). This model establishes the relationships between the output voltages of the Hall sensors and the distal spatial positions. By substituting Equation (6) into Equation (12), the representations of the actuator's spatial kinematics in both the actuator coordinate frame and the Cartesian coordinate frame can be obtained.

Interactive force model

Perception on interactive forces could be realized based on a statistical interaction model with inputs spatial kinematics and inner pressure. In the main axial actuation direction, based on our previous exploration,^{6,18} the relations of actuation force f , length l , and inner relative actuation pressure p are presented as follows:

$$f = -k(l - l_0) + ps, \quad (13)$$

where k is the structural stiffness of the actuator, decided by the structural design and material selection. $s = \pi r_0^2$ is the cross-sectional area. When an interaction force F applies, the actuator generates differences in spatial kinematics and inner pressure. As a result, the interactive force could be obtained

with sensory feedback of spatial kinematics and inner pressure. Therefore, the interaction force in linear motion is as follows:

$$F = Ml''(t) + G + f, \quad (14)$$

where M, G are the inertial matrix and gravity on the distal plate, and $l(t)$ represents the actuator length variation with time response. Substituting Equation (13) into (14), we have a complete representation of interaction force in linear motion, as follows:

$$F = Ml''(t) + G - k(l(t) - l_0) + ps. \quad (15)$$

Based on the spring-like feature of soft origami actuator, the interaction force F is composed of a component force F_1 vertical to the distal plate, a component force F_2 parallel to the distal plate, and a bending torque τ ,^{39,40} as shown in Figure 2c. Here, we focus on exploring the axial direction force that $F_1 = F \cos \alpha$. Therefore, the interactive force in the axial direction satisfies the following:

$$F = G + [Ml''(t) + ps - k(l(t) - l_0)] / \cos \alpha. \quad (16)$$

Equations (15) and (16) are used to measure the interactive force in the axial direction separately in linear and bending movements.

Workspace Verification

The validation of the SoMO design starts from the actuator motion range, focusing on the requirements for workspace matching between the actuator and the magnetic field. The multidimensional movements exhibited by soft origami actuators include contraction, extension, and omnidirectional bending in response to the internal actuation or external load. Consequently, the actuator workspace could be represented as follows:

$$\begin{cases} l \in [l_{\min}, l_{\max}] \\ \alpha \in [0, (l_{\max} - l_{\min}) / (2r_0)], \\ \beta \in [0, 2\pi] \end{cases} \quad (17)$$

where $l_{\max}$ and $l_{\min}$ are the maximum and minimum lengths of the actuator, respectively. The actuator could bend to the maximum extent that one side with maximum length $l_{\max}$ and the other side with minimum length $l_{\min}$. r_0 is the radius of actuator in the actuation state when the chamber inflates to approximate a cylinder. Equation (17) indicates that the complete motion range primarily relies on the maximum and minimum lengths. These values will be experimentally determined for the workspace verification.

Two types of actuators, featuring bellows and origami structures,^{6,32,33} were fabricated to demonstrate the extensibility of the proposed design. Both actuators showcased different motion ranges. They were fabricated with rubber materials (Rubber Hei-Cast 8400, shore hardness 80A) and had initial lengths l_0 of 36 mm and radii r_0 of 22.5 and 10 mm, respectively. An experimental platform for testing soft pneumatic actuators was constructed, shown in Figure 3a,

consisting of a stepper motor-driven slider offering linear movements, a laser sensor (*Panasonic*, HG-C1100, 50 mm $\pm$ 35 mm) for linear displacement recording, a force sensor (range: 0–50 N) mounting on the slider for linear force feedback, three pressure sensors (range: $\pm$ 100 kPa) connecting to three pneumatic circuits for positive/negative pressure feedback, and a microcontroller (*STM32 F103ZET6*). With this configuration, the actuation force f , actuator length l , and pressure p could be recorded in real time. Figure 3c and d shows the actuator length l variation under the increasing pressure p . In this test, the actuation force was kept at zero to ensure that

actuators work in free space. The results in Figure 3c show that the full travel range of the origami actuator fell into [17.4 mm, 42.8 mm]. The full motion range of the bellow actuator in Figure 3d is [10 mm, 49.7 mm]. By substituting these results into Equation (17), the workspaces of the bellow and origami actuators are separately illustrated in Figure 3e, f on the xoz plane. The full workspace is obtained by rotating around the z -axis.

Magnetic field workspace is primarily influenced by the dimensions of permanent magnets, in terms of the diameter and thickness in a cylindrical shape.⁴¹ In this study, the Hall

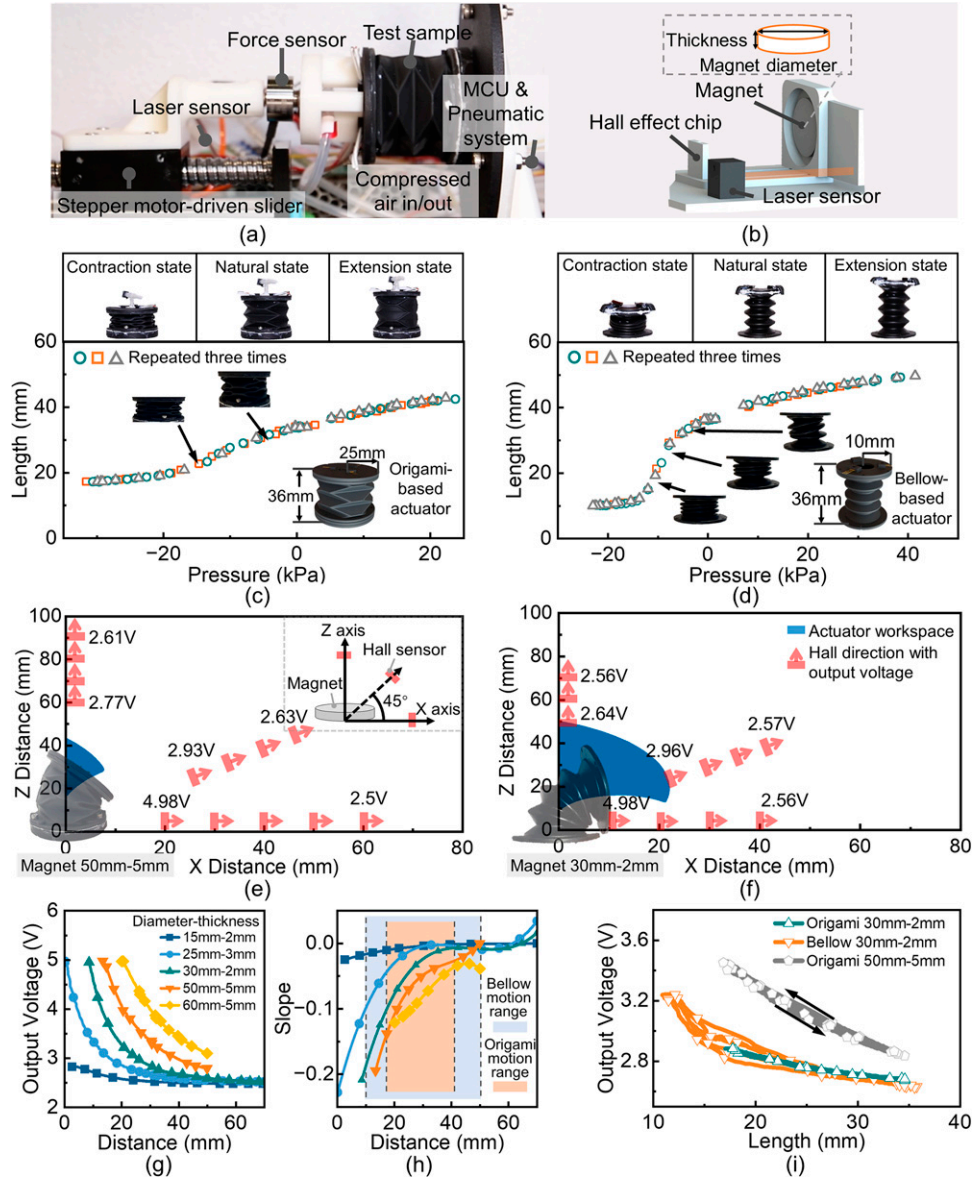


FIG. 3. Verification of workspace matching between the actuator and magnetic field. (a) Experimental platform for actuation performance tests and (b) magnetic field workspace tests. (c and d) Experimental results of the origami actuator and bellow actuator on relations of linear length and pressure in free space. (e and f) Comparison of workspace boundaries of far-field Hall magnetism and soft actuators (represented by the blue area, which indicates the trajectory of the distal plate center). (g) The output voltages following with distance variation between the Hall sensor and magnet for defining the space of the magnetic field. Five different sizes of magnets were tested. (h) Slope for the output voltages in relation to the distance of five tested magnets, in which actuator motion ranges are highlighted. (i) The relationship between output voltages and actuator length with Hall sensor inserted into the off-center of the distal plate.

sensor (EQ-730L, output: 0–5.0V, north pole output: 0–2.5V, south pole output: 2.5–5.0V, magnetic sensitivity: 130 mV/mT) and cylindrical Nd-Fe-B magnets in five different dimensions (diameter thickness: 15 mm–2 mm, 30 mm–2 mm, 25 mm–3 mm, 50 mm–5 mm, 60 mm–5 mm) were tested on a platform (Fig. 3b). The magnet was fixed to the platform, while the Hall sensor moved linearly along a slider. By changing the central distance between the magnet and sensor, the output voltages under the change of distances were recorded, as shown in Figure 3g. All these magnets could be used to measure the minimum and maximum lengths of soft actuators. As distances increase, the output voltages exhibit a nonlinear decrease from 5 to 2.5V, with significant signal variation occurring in the distance close to the magnet. This large-variation zone is more sensitive to the changes in distance. In other words, a larger slope value for the output voltages in relation to the distance would result in a higher resolution. We plot the slope of these magnets with actuator motion range as highlighted in Figure 3h. It suggests that magnets (30 mm–2 mm, 50 mm–5 mm, and 60 mm–5 mm) exhibit larger slope value within the actuator motion range. Another factor that needs to be considered is the design configuration of actuator. Consequently, to suit the dimension of actuator, magnets (30 mm–2 mm and 50 mm–5 mm) were selected for the bellow and origami actuators, respectively. In addition, by substituting the signal variations of Hall sensor with magnet (50 mm–5 mm) into Equation (12), we obtained the value of $c = 5 \times 10^{1.8}$. Following the magnetic field test procedure, we further tested the output voltages of Hall sensor by adding two other directions where the Hall sensor was positioned vertically along the x -axis, and the line inclined at a 45° angle to the x -axis. The output voltage variations in these directions verified the feasibility of Hall sensors in capturing the actuator workspace as given in Figure 3e, f. These tests were conducted to offer a preliminary verification for the conceptual design.

Finally, one Hall sensor was further integrated into the origami and bellow actuators. The position of magnet and Hall sensor is not center-to-center considering the specific configuration of soft actuation-sensing integrated design where the pneumatic tube is located in the center of distal plate. They were tested in the platform, as given in Figure 3a, to preliminarily validate the sensing performance of the SoMO actuator. In this test, the actuation force was kept at zero to ensure that actuators work in free space. The output voltages of one Hall sensor and actuation length were recorded as shown in Figure 3i. These loops also describe the low hysteresis of actuator, which is mainly affected by the actuator structure design and Hall signal reading frequency. The origami actuator (50 mm–5 mm) exhibited smaller hysteresis and better linearity than the bellow actuator (30 mm–2 mm). This superiority may result from the more linear motion of origami actuator instead of a slight unstable bending tendency during the contraction, as shown in the photographs in Figure 3d. Therefore, the origami structure is used to develop the SoMO actuator for better sensing performance. Consequently, the workspace matching between the actuator and magnetic field, as well as the feasibility of Hall sensor-integrated actuator design, was preliminarily validated.

Verification on Spatial Movement Perception

The SoMO actuator was fabricated, as illustrated in Figure 4a, to validate the sensing capabilities of measuring the spatial multi-dimensional movements. The fabrication proceeds in the following three steps: (1) fabricating the origami (bellow) chamber (Rubber Hei-Cast 8400, shore hardness 80A) by the conventional molding process; (2) inserting the magnets and Hall sensory array into the three-dimensional-printed bottom plate and distal plate, and (3) assembling the two plates with the chamber. In the SoMO actuator design, the dimensions of the chambers and magnets are consistent with the prototypes in the previous section. The parameters of the Hall sensor array are $r = 12.5$ mm and $n = 3$.

An experimental platform (Fig. 4b) was built to obtain the ground truth with an IMU (MPU6050, *InvenSense Inc.*) for recording the Euler angles of distal plate, and a laser sensor installed in the platform (repeatability $70 \mu\text{m}$, HG-C1100, *Panasonic Inc.*) for recording the distance between the laser sensor-installed position and actuator distal plate L . With this configuration, the spatial kinematics of actuator could be calculated through the following geometric relations Equation (17).

In this platform, real-time Euler angles, pitch p' , yaw y' , roll r' (angles rotating with x , y , and z axes), and central distance L , could be obtained. The central distance L is an important reference for the positioning of distal plate in the Cartesian coordinate system, as only orientation is provided by the IMU. The roll angle $r' = 0$ for the SoMO actuator. The normal vector of distal plate $\mathbf{n}$ in the Euler coordinate frame and the actuator coordinate frame is separately represented as follows:

$$\begin{cases} \mathbf{n} = (\sin y', -\cos y' \sin p', \cos p' \cos y')^T \\ \mathbf{n} = (\sin \alpha \cos \beta, \sin \alpha \sin \beta, \cos \alpha)^T \end{cases} \quad (18)$$

Therefore, according to (18), we could complete transformation between the Euler angles and actuator angles. Therefore, we have the following model translating IMU and laser sensor signals into actuator shape as ground truth:

$$\begin{cases} \alpha = \arccos(\cos p' \cos y') \\ \beta = \arctan 2(-\cos y' \sin p', \sin y') \\ l = (L_0 - L) \alpha / \tan \alpha \end{cases} \quad (19)$$

Consequently, the proposed platform could effectively illustrate the actuator movements as ground truth. Both the automatic actuation and manual operation are supported to be conducted with compensation and translation of model (19).

Perception of linear motion

Verification on spatial kinematics begins with linear motion. First, the actuator was linearly pressed, and released for one cycle in the platform (Fig. 4b). This cycle was repeated four times. Figure 4c shows the original signals read from the laser sensor and IMU, where the Euler angles from the IMU showed slight fluctuations. These fluctuations resulted from the manual operation, which were recorded and considered in the length calculation based on the model (19). The length configurations of the origami and bellow

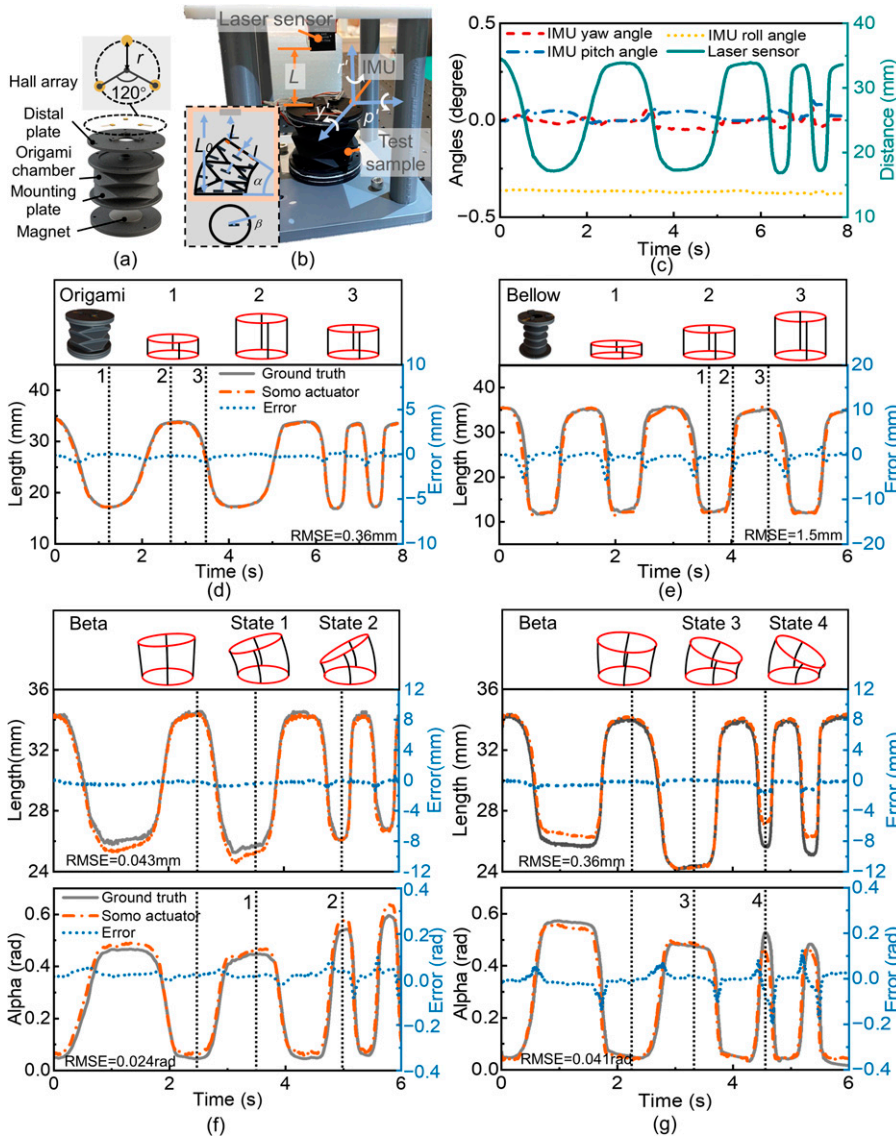


FIG. 4. Verification of spatial multidimensional movement perception. (a) Exploded view of Soft Far-Field Magnetic Origami (SoMO) actuator. (b) Experimental platform for capturing the ground truth in linear motion and omnidirectional bending motion. (c) Original signals from Inertial Measurement Unit (IMU) and laser sensor as the ground truth. (d, e) Verification of linear motion of origami actuator and bellow actuator under press. The press was repeated four times. (f, g) Verification of bending motion of SoMO actuator in different directions. The actuator was manually bent in two directions for States 1 and 2 and States 3 and 4.

actuators are shown in Figure 4d, e, respectively. In this figure, both the ground truth calculated from commercial sensors (IMU and laser sensor) and the perceptive information from the Hall sensory array decoded by the spatial kinematic model (12) are depicted to verify the accuracy of the proposed approach. It is indicated that the measured information could be used to represent the actuator length l with origami root mean square error (RMSE) = 0.36 mm and bellow RMSE = 1.50 mm. By comparing the values of RMSE, the origami actuator showed better sensing performance than the bellow actuator. A slight deviation between ground truth and perception value was observed, possibly caused by the unstable contraction of the actuator, resulting in the slight shifting of Hall sensors.

Perception of omnidirectional bending motion

Bending motion was carried out on the experimental platform (Fig. 4b) to further validate the sensing performance in multidimensional movements. The SoMO actuator was manually pressed in a series of random β . Figure 4f, g plots the kinematic variation in the actuator coordinate frame under

passive pressure. Here, the ground truth was obtained by the model (19) based on the signals from laser sensor and IMU. The measured movements were obtained by the spatial kinematic model Equation (12) in the actuator coordinate frame based on Hall sensory array signals. The results show that the actuator could capture the change of spatial kinematics. Slight deviations were shown in the large bending angle range, where the actuator was largely bent. This conclusion was supported by the results of States 1 and 2 (Fig. 4f). The actuator in State 1 completed bending angle α from 0.047 rad to 0.446 rad with RMSE = 0.024 rad, orientation angle β (2.69 rad of perception and 2.92 rad of ground truth) with error = 0.23 rad, length l from 34.49 mm to 25.84 mm with RMSE = 0.43 mm, while the actuator in State 2 was bent from $\alpha = 0.058$ rad to 0.60 rad with RMSE = 0.041 rad, orientation angle β (2.82 rad of perception and 2.75 rad of ground truth) with error = 0.07 rad, length l from 34.38 mm to 26.18 mm with RMSE = 0.36 mm. By comparing the values of RMSE, we have the conclusion that the proposed actuator could achieve considerable accuracy in spatial bending movement perception, thus verifying the sensing

capabilities in bending movements. The actuator in States 3 and 4 was bent in the other direction, as shown in Figure 4g, that is, different orientation angle β . In State 3, it was bent from $\alpha = 0.05$ rad to 0.57 rad with RMSE = 0.028 rad, orientation angle β (0.15 rad of perception and 0.14 rad of ground truth) with error = 0.01 rad, length l from 34.12 mm to 24.28 mm with RMSE = 0.42 mm. In State 4, it was bent from $\alpha = 0.04$ rad to 0.46 rad with RMSE = 0.05 rad, orientation angle β (0.16 rad of perception and 0.14 rad of ground truth) with error = 0.02 rad, length l from 33.99 mm to 23.09 mm with RMSE = 0.75 mm.

The above experiments have verified the capability of multi-dimensional bending movement recognition in multidirections. Values of RMSE under various shapes suggest that high accuracy was achieved in linear motion. The accuracy was decreased when the actuator bent in a larger bending angle, which might be attributed to the extreme stretching and compression of two sides, thus introducing slight deviation between the model predictions and the actual behavior of the actuator.

Verification on Interactive Force Perception

Actuation force of SoMO actuator

First, we validate the actuation force of soft origami actuator, as modeled in Equation (13). In this equation, there are

parameters of structural stiffness k and effective cross-sectional area s obtained by experiments. During the test, the relationship between the actuation force f and pressure p was investigated with fix length l in the experimental platform (Fig. 3a). First, the actuator length l was fixed to 31 mm. We recorded the actuation force f under increasing pressure p , as depicted in Figure 5a. These results, together with the relationship between length and pressure under $f = 0$ tested in Figure 2c, were substituted into Equation (13) to get $k = 3062$ N/m and $s = 1.6 \times 10^{-3}$ m². Subsequently, simulation results of model Equation (13) in $l = 28$ mm are added to the figure to verify the proposed model. Agreements between experimental and modeled results were observed with RMSE = 0.78 N in $l = 31$ mm and 0.47 N in $l = 28$ mm, showing the actuation performance of the origami actuator.

Interactive force perception in axial direction

The capabilities of sensing the interactive force in the axial direction were validated in the spatial linear and bending movements. The interactive forces were given, as illustrated in Figure 5a, b, where a force sensor is used to record the applied force as the ground truth. This force was measured by substituting the recorded time-varying pressure and

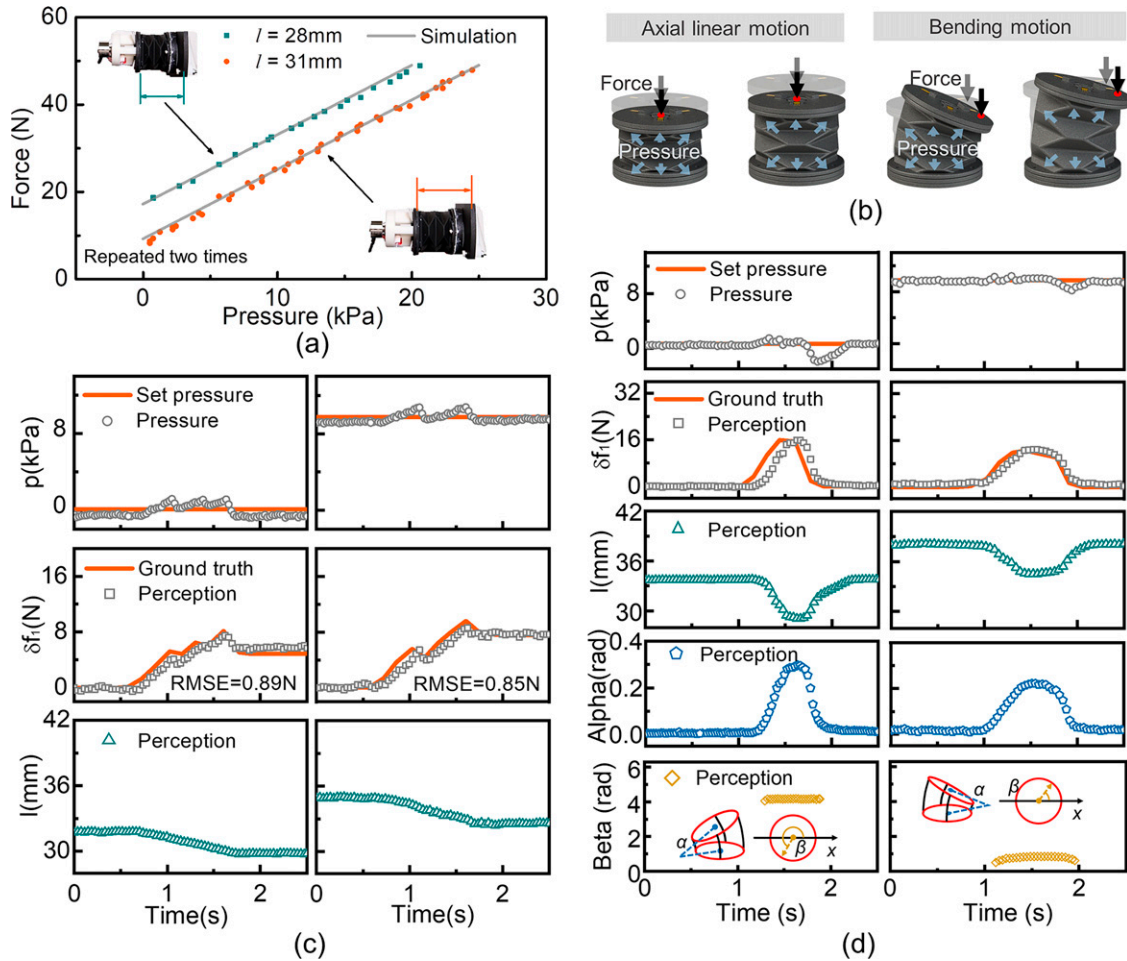


FIG. 5. Verification of interactive force perception. (a) Experimental results on relations of linear force and pressure in fixed lengths. (b) Illustrations of force tests separately in linear motion and bending motion. (c) Interactive forces of axial direction in linear motion with two sets of different actuation pressures. (d) Interactive forces of axial direction in bending motion with two sets of different actuation pressures.

spatial kinematics into Equations (15) and (16). Experiments were conducted on the experimental platform illustrated in Figure 3a. During the test, actuator pressure p was maintained constant. The interactive forces F_1 were applied by the motor-driven slider with designated displacements, leading to changes in actuator length. The variation of interactive forces in linear motion was recorded where two separate groups with actuator pressures of 0 kPa and 10 kPa were tested. Both groups experienced 3 mm slider displacements to compress the actuator. The results indicate that the interactive forces of the SoMO actuator increase as the actuator length l decreases. It was observed that the soft actuator's internal pressure fluctuates during this process, causing similar force variations. These values were input into Equation (15), with both the inertial matrix and gravity on the distal plate equated to zero due to the absence of an external load. The measured force variation, ground truth, and linear length variation were all plotted in Figure 5c. The perceived results and ground truth exhibited a consistent trend, with RMSE values of 0.89 and 0.84N.

To further verify the developed model in spatial bending movements, we demonstrated the variations of length l , bending angle α , and orientation angle β in Figure 5d. It should be noticed that a singularity occurs when α fluctuates within a small range for calculation of orientation angle β . Hence, to ensure an accurate tracking, the orientation angle β was calculated only when $\alpha > 0.05$ rad. By substituting the variation values of l and α into Equation (16), we have the axial force variation in bending movements, shown in

Figure 5d. The results support the conclusion that the developed model can effectively sense axial interactive forces.

Demonstration on Soft Robotic System

Control performance of the actuation system

Spatial movement and force perception offers vital pre-conditions for system performance control, including position feedback control and interactive force tuning. In this section, a demonstration on preliminary feedback control performance is presented.

The position control and force tuning strategies were implemented in a pneumatic system, which consists of eight solenoid valves (frequency: 30 Hz max.) and pressure sensors (± 100 kPa) providing positive and negative pressure control for four circuits with the aid of a microcontroller (STM32 F103ZET6). The gas supply is secured by two micropumps (420 L/h, max. 0.2 MPa–0.075 MPa). With this configuration, classical PD controllers with position feedback control (Fig. 6a) and pressure feedback control (Fig. 6b) were proposed to realize real-time position control and interactive force tuning, respectively. The interactive force-tuning strategy follows as when external force works on the soft robots and produces a disturbance, this disturbance is translated into the change of spatial kinematics, and further considered in the force tuning controller with a compensated pressure to actively change the interactive forces.

First, step response tests were conducted to demonstrate the performance of position control. Results on steps with

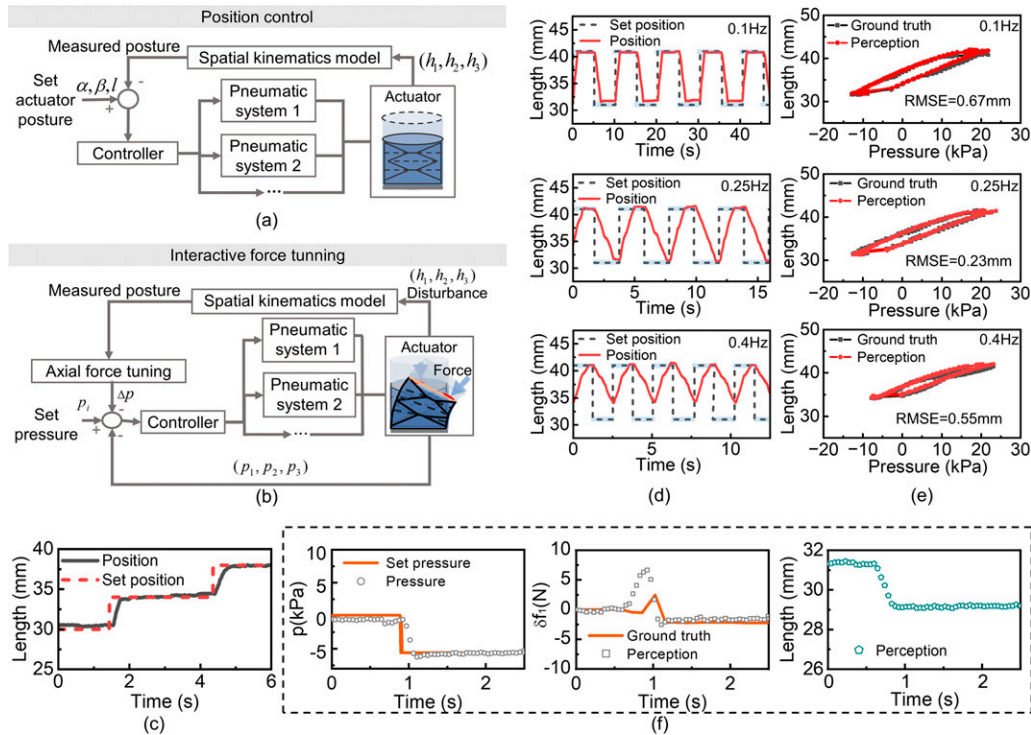


FIG. 6. Demonstrations of feedback control performance in the Soft Far-Field Magnetic Origami (SoMO) actuator. (a) Schematic diagrams for position feedback control and (b) pressure feedback for force tuning. (c) Step response in linear motion. (d) Position tracking following square-wave functions with frequencies of 0.1 Hz, 0.25 Hz, and 0.4 Hz. (e) Comparison of actuation and sensing system performance with respect to hysteresis. (f) Interactive force tuning demonstration. The sample size for calculating root mean square error (RMSE) is approximately 20 per second.

12 and 4 mm are separately shown in Figure 6c, demonstrating a fast response within 500 ms to the target position and high accuracy within 0.2 mm. The performance of position tracking following the square-wave functions with frequencies of 0.1, 0.25, and 0.4 Hz is shown in Figure 6d. The results demonstrate the high repeatability in real-time position tracking. The proposed soft robotic pneumatic system could fully catch the 0.25 Hz square wave. The relations of

length and pressure in these tests are also plotted in Figure 6e, showing the hysteresis of actuation systems. When the frequency rises to 0.4 Hz, the actuation system is insufficient in giving negative pressure to get the target position. However, the sensory system is always capable of capturing the position variation in these frequencies. This conclusion is supported by plotting the relations of length and pressure in these tests in Figure 6e. Here, the hysteresis

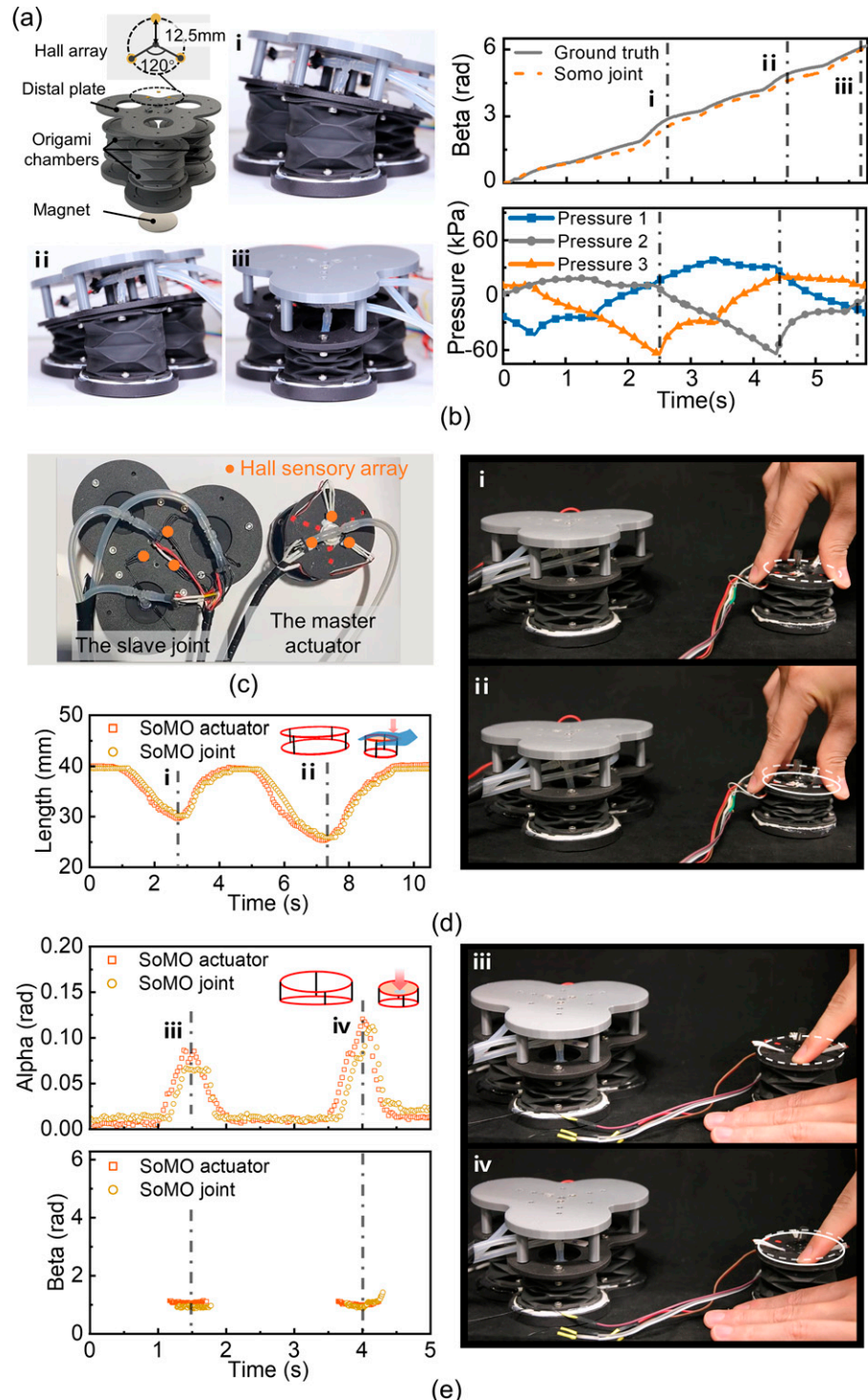


FIG. 7. Demonstrations of the Soft Far-Field Magnetic Origami (SoMO) actuator and joint system. (a) Exploded view of the SoMO joint. (b) Spatial orientation tracking of the SoMO joint, with the upper graph showing time-varying orientation and the lower graph displaying the actuation pressure applied to the joint. (c) Prototypes of the master actuator and slave joint. (d and e) Linear and bending motion tracking for the SoMO actuator and joint.

loops of the Hall array and ground truth from laser sensors are highly overlapped with each other, with RMSE of 0.67, 0.23, and 0.55 mm, respectively. It suggests that the hysteresis mainly resulted from the actuation system, not the sensing system, further verifying the sensing capabilities in dynamic systems.

Interactive force tuning capabilities are enabled by regulating the actuation pressure during the interaction. Different from the process of validating the perception of interactive force in the previous section where internal pressure was set at a constant, the force tuning process involves active pressure regulation in the robotic system to increase or decrease the interactive force. In this test, an axial force was applied to push the distal plate by the motor-driven slider. Figure 6f illustrates the process of regulating pressure and shows the variation of interactive force. The intensities of interactive forces have been effectively tuned by decreasing the inner actuation pressure, showing a similar tendency to the ground truth. It is observed that there is a sharp increase in the early stage of the interaction, resulting in an apparent deviation between the ground truth and measured force. This deviation was possibly caused by a sudden pressure increase under the regulation of solenoid valves with on-off operation modes.

Hence, the frequency of the solenoid valves is crucial for the response of the soft actuators. The maximum frequency of solenoid valves used in the developed actuation system is 30 Hz. Utilizing solenoid valves with higher frequency would be helpful for the higher actuation performance.

Demonstration on SoMO robotic joint

To explore the extendibility of the SoMO design and sensing mechanism, a SoMO robotic joint was developed, as shown in Figure 7a. In this joint, three origami chambers are arranged in parallel, connected by a mounting plate and a distal plate. This arrangement allows the proposed joint to exhibit omnidirectional bending movements under both internal actuation and external interaction. Following the design principle of the Hall sensory array, the simplest configuration to capture the distal position and body shape is an equilateral triangle array. Therefore, the Hall sensory array and magnet are utilized in the same manner as the SoMO actuator. Specifically, three Hall sensors are placed in the distal plate with $r = 12.5$ mm; and a magnet is inserted into the mounting plate with dimensions of 50 mm-5 mm. Demonstrations of continuous rotation with a variation of orientations were performed on the SoMO joint by giving a series

TABLE 1. SUMMARY OF PARAMETERS

Classification	Symbols	Physical meanings
Design parameters	r_0	Radius of soft pneumatic actuator
	l_0	Actuator initial lengths
	r	Distance of Hall sensor inserting point to the center of distal plate
	n	Number of Hall sensors in the sensory array
	$\theta_{i,i=1,2,3}$	Mounting angle from the x_0 -axis to the Hall sensor
	h_0	Initial signals of Hall sensors
	σ	Radius of a cylindrical magnet
	d	Thickness of a cylindrical magnet
	k	Structural stiffness of the actuator
	s	Cross-sectional area of actuator
Magnetic field-related parameters	ϕ	Magnetic flux
	c	Constant related to the Hall sensor specification
	$\mathbf{B}$	Magnetic flux density
	$\mathbf{n}$	Unit normal vector connecting to the orientation of distal plate
	μ_T	Relative permeability of the air
	μ_0	Air magnetic permeability
	m_T	Magnetic intensity of magnet
	m_0	Magnetization strength for an Nd-Fe-B magnet.
	$h_{i,i=1,2,3}$	Output signals of three evenly distributed Hall sensors
	p	Actuation pressure
Actuator posture in actuator coordinate frame	α	Bending angle of actuator
	β	Orientation angle of distal plate from the x_0 -axis to the bending position
	l	Central length of actuator
	$l_{i,i=1,2,3}$	Lengths of sensor lines in parallel with central line
Actuator coordinates in Cartesian coordinate frame	$H_{i,i=1,2,3}(x_i, y_i, z_i)$	Spatial positions of points inserting Hall effect chips
Actuator force parameters	f	Actuation axial force
	F	Interaction force
	F_1	Component force of F vertical to the distal plate
	F_2	Component force of F parallel to the distal plate
	M	Inertial matrix on the distal plate
	G	Gravity on the distal plate
Parameters used in the experimental platform	L	Distance between the laser sensor-installed position and actuator distal plate
	p'	Pitch Euler angle
	y'	Yaw Euler angle
	r'	Roll Euler angle

of pressures to these three origami chambers. In this test, an IMU was attached to the distal plate for collecting ground truth information. The actuator rotation in free space is represented in Figure 7b, where three chambers were pressurized in sequence, ranging from -60 to 20 kPa. The Hall sensory array signals were processed with the same spatial kinematic model as the SoMO actuator. The results indicate that the SoMO joint is able to complete full rotation from $\beta = 0$ to $\beta = 2\pi$ rad, and demonstrate consistency between the ground truth and the estimated orientation.

A soft master–slave robotic system was established, as in Figure 7c, where the SoMO actuator acts as the master system to receive the external interaction and the joint acts as the slave system. In this demonstration, the stimuli on the master actuator were applied by a human hand, resulting in a linear or bending motion in the master actuator. The motion was measured by the Hall sensory array. When the deviation of length and bending angle exceeded the dead zone ± 1 mm of length or 0.03 rad of bending angle from the initial state, the slave joint responded with the same motion. This process was repeated twice. In the linear motion tests, the delay time for initiating motion between the master actuator and slave joint was 0.66 and 0.15 s in these two processes, as illustrated in Figure 7d. Several factors, such as the speed of the human hand on the master–slave and the range of the dead zone, can impact the delay time. The proposed robotic system exhibited distinct delay times in these two processes due to the relatively slow speed of the human hand within the deviation zone, leading to a slower response in the first process. Nevertheless, high accuracy in following the peak position was observed with errors of 0.70 mm and 0.69 mm, and time intervals of 0.06 s and 0.05 s. The results of the bending motion tests also emphasize the same point as shown in Figure 7e, with accurate tracking of both the orientation angle β and bend angle α . The peak bending angle errors in these two repeated processes were 0.02 and 0.01 rad, with time intervals of 0.06 and 0.13 s.

Conclusion and Future Work

In this study, we investigated the potential of utilizing far-field magnetic sensing combined with a soft actuator model-based approach for multimodal perception. We introduced the SoMO design, which incorporates a compact triangularly arranged Hall sensory array, to accurately track the distal position of a soft pneumatic origami actuator. By leveraging the modeled relationships between the nonstrain measurable parameters of soft origami actuators (distal position and pressure) and the desired sensing modalities, we successfully retrieved spatial multidimensional movement and force information. We validated the workspace matching between the actuator and the magnetic field and sensing performance in spatial multidimensional movements (linear and omnidirectional bending motions) (a summary of all parameters used in this study is listed in Table 1). The SoMO actuator demonstrated accurate spatial kinematic perception with a root-mean-square deviation of 0.36 mm in length and 0.02 rad in orientation, as well as interactive force variation detection with root-mean-square deviation of 0.89 N. In addition, it detects interactive force variations with a root-mean-square deviation of 0.89 N (a summary of the performance of SoMO approach and state of the art is listed in Table 2).

TABLE 2. PERFORMANCE SUMMARY OF THE SOFT FAR-FIELD MAGNETIC ORIGAMI APPROACH AND THE STATE OF THE ART

State of the art	Sensing mechanism	Soft actuators type	Measured parameters	Sensing modality			Performance (error)		
				Linear	Bending	Force	Displacement	Angle	Force
Zhao ²⁰	Optics	Hydraulic fiber-reinforced actuators	Strain	✓	✓	✓	—	—	—
Luo ²⁴	Magnetics	Pneumatic fiber-reinforced actuators	Strain	—	✓	—	—	—	—
Wirekoh ²⁹ Shen ³² Scharff ³³ Tiziani ³⁴ Yang ³⁵	Resistance	Pneumatic artificial muscles	Strain	✓	—	✓	~ 0.06 mm (full range 2.7 mm)	—	0.68 N
	Optics	Pneumatic origami actuator	Strain	✓	✓	—	—	0.04 rad	—
	Optics	Pneumatic bellows	Distal position	✓	✓	—	—	—	—
	Optics, pressure	Pneumatic artificial muscle	Distal position	✓	—	✓	1.3 cm (full range ~ 70 cm)	—	0.16 N
	Optics, Resistance	Pneumatic artificial muscle	Distal position, pressure	✓	—	✓	—	—	—
Sundaram ³⁶	Far-field magnetics	Electro-hydraulic actuators	Distal position	✓	✓	—	± 0.15 mm (full range 12 mm)	—	—
This work	Far-field magnetics, pressure	Pneumatic origami actuators	Distal position, pressure	✓	✓	✓	0.36 mm (full range 17 mm)	0.02 rad	0.89 N

Leveraging this high sensing performance for feedback control, we developed the soft robotic SoMO actuator and joint for position control and interactive force tuning. The actuator's capabilities were demonstrated through step response and square wave function following, exhibiting a fast response time of 500 ms for precise control with a 0.2 mm accuracy and the ability to track a 0.25 Hz square wave. In addition, the proposed approach was validated by exhibiting highly overlapping hysteresis loops of sensory information in conjunction with the actuation system, resulting in an RMSE of 0.55 mm for a 0.4 Hz square wave. This further verifies the sensing capabilities of the system in dynamic scenarios. Moreover, the interactive force perception facilitated force tuning during interaction by regulating inner pressure. Finally, this complete methodology was demonstrated in the SoMO joint and actuator, which were contributed to a soft master-slave robotic system. This showcased the extendibility of the proposed approach and its capabilities for perceiving and responding to a soft robotic system. Beyond these highlighted results, the methodology presented in this work serves as a guidance for soft robotic perception, covering conceptual design, fabrication, and modeling for multidimensional deformation, as well as strategies for interactive force tuning.

One primary limitation of this approach is the susceptibility of the Hall sensor array to interference from electromagnetic noise, which can affect sensing precision and consequently limit certain applications of the proposed method. Future work will investigate potential solutions for immunization against such interference. Another limitation is the use of piecewise constant curvature for simplification of modeling at the expense of reduced accuracy. More precise models will be explored in future research. Moreover, the potential for spatial multidimensional kinematics and interaction perception in the perception and control of a multisegment robotic system will be explored, including spatial position control and force perception in other directions for broader interaction applications.

Authors' Contributions

J. Yi led the conceptualization, experimental design, data analysis, and manuscript drafting; J. Xu designed experiments and was responsible for the visualization of figures and videos; Y. Deng, Y. Xuan, and Z. Fang participated in concept validation; B. Lou took part in the experimental implementation; K. Zou, Y. Su, and Q. Tan contributed to figure visualization; R. Wen, S. Liu, and Z. Wang reviewed the manuscript and provided input on the final draft.

Author Disclosure Statement

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Supplementary Material

Supplementary Video

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